

Practicing and Extending Joins

DATA 202 21FA

Reminders

- DS Firepit Tonight!
- Quiz 5 closes tonight
- Homework 3
- Discussion Forum replies

Preview: replication project

Gradescope trial

See Moodle.

Relational Data

- Bring together data that may never have been brought together before
- Community, interdependence

Example Applications

- Looking up abbreviations
- Connecting sales data to each customer's demographics
- Combining student data from Moodle, an online textbook, GitHub activity reports, ...

What others can you think of?

Code Together: Flight Delays

```
library(nycflights13)
```

```
flights %>%  
  drop_na(arr_delay) %>%  
  group_by(carrier) %>%  
  summarize(avg_delay = mean(arr_delay)) %>%  
  arrange(desc(avg_delay)) %>%  
  left_join(airlines, by = "carrier") %>%  
  select(name, avg_delay)
```

```
# A tibble: 16 × 2
```

	name	avg_delay
	<chr>	<dbl>
1	Frontier Airlines Inc.	21.9
2	AirTran Airways Corporation	20.1
3	ExpressJet Airlines Inc.	15.8
4	Mesa Airlines Inc.	15.6
5	SkyWest Airlines Inc.	11.9
6	Envoy Air	10.8
#	... with 10 more rows	

Multiple matches

`left_join(x, y)`

1	x1	1	y1
2	x2	2	y2
3	x3	4	y4
		2	y5

Revenue by item?

purchases

customer_id	item
c1	bread
c1	milk
c1	banana
c2	milk
c2	toilet paper

prices

item	price
avocado	0.50
banana	0.15
bread	1.00
milk	0.80
toilet paper	3.00

Revenue by item?

For each item, look up all sales data.

```
prices %>%  
  left_join(purchases)
```

```
# A tibble: 6 × 3  
  item      price customer_id  
  <chr>    <dbl>   <chr>  
1 avocado    0.5    <NA>  
2 banana    0.15    c1  
3 bread      1      c1  
4 milk      0.8    c1  
5 milk      0.8    c2  
6 toilet paper 3      c2
```

Notice: multiple rows for each item. Where did each one come from?

Next: count how many purchases for each item.

Revenue by item?

Count how many purchases for each item.

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item) %>%  
  summarize(count = n())
```

```
# A tibble: 5 × 2  
  item      count  
  <chr>    <int>  
1 avocado      1  
2 banana      1  
3 bread       1  
4 milk        2  
5 toilet paper 1
```

Revenue by item?

Count how many purchases for each item.

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item) %>%  
  summarize(count = n())
```

```
# A tibble: 5 × 2  
  item      count  
  <chr>    <int>  
1 avocado      1  
2 banana      1  
3 bread       1  
4 milk        2  
5 toilet paper 1
```

Could have summarized `purchases` first. Advantages to that approach? This one?

Revenue by item?

Count how many purchases for each item.

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item) %>%  
  summarize(count = n())
```

```
# A tibble: 5 × 2  
  item          count  
  <chr>        <int>  
1 avocado          1  
2 banana           1  
3 bread            1  
4 milk             2  
5 toilet paper     1
```

Could have summarized `purchases` first. Advantages to that approach? This one?

Revenue by item?

Thread the prices along. (Check: any extra rows?)

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item, price) %>%  
  summarize(count = n())
```

```
# A tibble: 5 × 3  
# Groups:   item [5]  
  item           price count  
  <chr>         <dbl> <int>  
1 avocado         0.5      1  
2 banana         0.15     1  
3 bread           1       1  
4 milk           0.8      2  
5 toilet paper    3       1
```

What next?

Revenue by item?

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item, price) %>%  
  summarize(count = n(), revenue = price * count)
```

```
# A tibble: 6 × 4  
# Groups:   item, price [5]  
  item          price count revenue  
  <chr>        <dbl> <int>   <dbl>  
1 avocado      0.5      1     0.5  
2 banana      0.15     1    0.15  
3 bread        1        1     1  
4 milk        0.8        2    1.6  
5 milk        0.8        2    1.6  
6 toilet paper 3         1     3
```

Hmm... where did the multiple rows come from?

Revenue by item?

Sum the prices, *weighted* by count.

```
prices %>%  
  left_join(purchases) %>%  
  group_by(item, price) %>%  
  summarize(count = n(), revenue = sum(price * count))
```

```
# A tibble: 5 × 4  
# Groups:   item [5]  
  item      price count revenue  
  <chr>    <dbl> <int>   <dbl>  
1 avocado    0.5      1     0.5  
2 banana    0.15      1    0.15  
3 bread       1      1     1  
4 milk       0.8      2    3.2  
5 toilet paper 3      1     3
```


uh oh...

customer_id	item
c1	bread
c1	milk
c1	bananas
c2	milk
c2	toilet paper

item	price
AVOCADO	0.50
BANANA	0.15
BREAD	1.00
MILK	0.80
TOILET_PAPER	3.00

uh oh...

customer_id	item
c1	bread
c1	milk
c1	bananas
c2	milk
c2	toilet paper

item	price
AVOCADO	0.50
BANANA	0.15
BREAD	1.00
MILK	0.80
TOILET_PAPER	3.00

```
purchases %>%  
  left_join(prices)
```

```
# A tibble: 5 × 3  
  customer_id item      price  
  <chr>      <chr>    <dbl>  
1 c1        bread      NA  
2 c1        milk       NA  
3 c1        bananas    NA  
4 c2        milk       NA  
5 c2        toilet paper NA
```

Specifying keys

- Keys must match *exactly*
- Can join on multiple columns (first name **and** last name)
- Default join: columns with same names
- Specify what columns to use: `left_join(x, y, by = c("first_name", "last_name"))`